X. Appendix

A. Related Work: Extended Survey

Privacy-Preserving Techniques in FL. Gradient inversion attacks [2] show that an honest-but-curious server can reconstruct raw samples from shared gradients. DP-SGD [3] clips gradients and injects calibrated noise for client-level (ϵ, δ)-DP. Secure aggregation [14] uses cryptographic masking so the server sees only the aggregate. Both treat ϵ as a fixed parameter, not a traded commodity.

FL Incentive Mechanisms. Coalitional games [39], Stackelberg games [15], [16], MFG-based mechanisms [12], and matching games [17] address incentive design. AFL [19] is an active sub-field; CSRA [11] provides robust reverse auctions for DP-FL. All existing AFL works trade data, models, or compute rather than privacy budgets.

Privacy Auctions. Ghosh & Roth [8] laid the theoretical foundation; Fleischer & Lyu [10] extended to correlated costs; Ghosh et al. [20] handled unverifiable reports; Wang et al. [40] studied negative payments; Zhang et al. [28] addressed auctions under privacy constraints; Hu et al. [13] proposed MFG-based data pricing; Wu et al. [21] is the closest prior art without a learned differentiable auction. All prior works are static, single-round, and centralised.

Differentiable Mechanism Design. Myerson [7] gives revenue-optimal auctions under restrictive assumptions. RegretNet [9] parameterises allocation/payment rules as neural networks trained to minimise ex-post regret. DM-RegretNet [5] applies this to FL model trading but inherits $\mathcal{O}(N^2)$ pairwise dependency. MFG-RegretNet reduces this to $\mathcal{O}(N)$ and operates on privacy budgets as the commodity.

B. Formal Preliminaries

Definition 7 (PLDP [25]) *For user i with $\epsilon_i{>}0, \delta_i{\geq}0$, mechanism $\mathcal{M}^i$ is (ϵ_i, δ_i)-PLDP if for all adjacent inputs $\boldsymbol{x}, \boldsymbol{x}'$ and outputs $\boldsymbol{y}$:* $\Pr[\mathcal{M}^i(\boldsymbol{x}){=}\boldsymbol{y}] \leq e^{\epsilon_i} \Pr[\mathcal{M}^i(\boldsymbol{x}'){=}\boldsymbol{y}] + \delta_i$.

FL global model: $\mathbf{w}_G{=}\sum_i \alpha_i \mathbf{w}_i$, $\mathbf{w}_G^*{=}\arg\min_{\mathbf{w}} \sum_i \alpha_i \mathcal{L}_i(\mathbf{w}_i)$. Gradient ℓ_2-sensitivity: $\Delta f_{\mathbf{w}_i} = \max_{\mathcal{D}_i \sim \mathcal{D}_i'} \|\nabla \mathcal{L}_i(\mathbf{w}, \mathcal{D}_i) - \nabla \mathcal{L}_i(\mathbf{w}, \mathcal{D}_i')\|_2$.

C. HJB-Fokker-Planck System

The mean-field bid statistic is $b_{\mathrm{MFG}}(t) = \mathbb{E}_{\mu_t}[b] = \int_{\mathcal{S}} b(s; \mu_t)\, d\mu_t(s)$. The value function $\Phi(s, t; \mu)$ satisfies the **HJB** equation:

$$-\partial_t \Phi - \mathcal{H}(s, \nabla_s \Phi, \mu_t) - \tfrac{\sigma^2}{2} \Delta_s \Phi = 0, \quad \Phi(s, T) = 0, \quad (43)$$

with Hamiltonian $\mathcal{H} = \max_b [\tilde{p}(b; \mu_t) - c(v, \epsilon(b)) + (\nabla_s \Phi)^\top f(s, b, \mu_t)]$ and mean-field payment $\tilde{p}(b; \mu_t) = \mathbb{E}[p(b, B_{-i})]$. The population density evolves via the **Fokker-Planck** equation:

$$\partial_t \mu_t + \nabla_s \cdot (\mu_t f(s, b^*, \mu_t)) = \tfrac{\sigma^2}{2} \Delta_s \mu_t, \quad \mu_0 = \mathcal{W}. \quad (44)$$

Equations (43) and (44) form a coupled forward-backward system whose fixed point is the MFE [41].

D. Value-Function Baselining

Proposition 1 (Variance Reduction via Value Baselining) *Let $X_i^{(r)}{:=}u_i(b_i'^{(r)}, b_{-i}; b'_{\mathrm{MFG}}, \theta) - u_i(b_i, b_{-i}; b_{\mathrm{MFG}}, \theta)$. For any baseline c independent of b_i', $\mathbb{E}[\hat{X}_i^{(r)}]{=}\mathbb{E}[X_i^{(r)}]$ (unbiased). The optimal baseline $c^*{=}\widehat{\Phi}_\xi(b_i, b_{\mathrm{MFG}})$ minimises $\mathrm{Var}(\hat{X}_i^{(r)})$, with reduction $\mathrm{Var}(\widehat{\Phi}_\xi)/\mathrm{Var}(X_i^{(r)})$.*

The critic $\widehat{\Phi}_\xi$ is trained via HJB residual minimisation: $\mathcal{L}_{\mathrm{HJB}}(\xi) = \mathbb{E}[(\widehat{\Phi}_\xi - \mathcal{T}\widehat{\Phi}_\xi)^2]$, where $\mathcal{T}\widehat{\Phi}$ is a one-step Bellman target from $\tilde{p}$. The critic shapes the regret penalty only; payments remain $\bar{p}_i$ from the Payment Network.

E. Proofs

Theorem 1 (Privacy Auction Game Equilibrium) *In a PAG where privacy valuations v_i of DOs are drawn from joint distribution $\mathcal{W}$, the cost function is linear $c(v_i, \epsilon_i) = \epsilon_i \cdot v_i$, and the mechanism $\mathcal{M} = (q, p)$ satisfies DSIC, IR, DT, and BF, there exists a Bayesian-Nash Equilibrium in which all rational bidders truthfully report both their privacy valuations v_i* and *their privacy budgets ϵ_i:*

$$\forall i \in \mathcal{N}, \forall v_i \in \mathcal{V}_i, \quad \mathbb{E}[u_i(q, v_i)] \geq \mathbb{E}[u_i(q_{-i}, v_i)]. \quad (45)$$

Proof Sketch: Consider the threshold-based PAC mechanism (Algorithm 1), where valuations are sorted $v_1 \leq \cdots \leq v_n$, the largest k is selected such that $c(v_k, 1/(n{-}k)) \leq B/k$, and payments are $p_i = \min(B/k, c(v_{k+1}, 1/(n-k)))$ for $i \leq k$ and $p_i = 0$ otherwise.

DSIC (for $i \leq k$): Misreporting a higher v_i' may cause i to drop out of the top-k, yielding $p_i' = 0$ and $u_i' = 0$. Since $u_i = p_i - c(v_i, \epsilon_i) \geq 0$ (by IR), deviation yields no gain.

DSIC (for $i > k$): Misreporting a lower v_i' may enter the top-k, but the diluted payment $p_i' \leq B/(k{+}1)$ combined with the higher true cost $c(v_i, \epsilon_i') > c(v_i', \epsilon_i')$ results in $u_i' < 0$ or no improvement.

By monotonicity of the mechanism, misreporting cannot improve the allocation-payment tradeoff, so truthful reporting is a dominant strategy (DSIC) and all utilities are non-negative (IR).

BNE: By DSIC, for any fixed b_{-i}, truthful reporting maximises $u_i(v_i, \cdot)$. Taking expectations over $v_{-i} \sim \mathcal{W}_{-i}$ yields Eq. (45), which is the BNE definition.

For finite N, existence follows from Nash's theorem. In the large-N regime, the multi-agent game converges to a Mean-Field Game, where fixed-point theorems from MFG literature [41] guarantee equilibrium existence. ■

Theorem 2 (Existence of MFE) *Under Assumption 2, the privacy auction MFG admits at least one Mean-Field Equilibrium (Φ^*, μ^*).*

Proof: We apply Schauder's fixed-point theorem. Let $\mathcal{M} = \mathcal{C}([0, T]; \mathcal{P}_2(\mathcal{S}))$ with the sup-Wasserstein-1 metric. Define $\mathcal{T} : \mathcal{M} \to \mathcal{M}$ by $\mathcal{T}(\mu) = \hat{\mu}$, where $\hat{\mu}_t$ is the law of $\mathbf{s}(t)$ under the SDE of Eq. (18) driven by the optimal $b^*(\cdot, t; \mu)$ from Eq. (43). *Well-definedness* follows from viscosity solution theory under (R1) through (R3) [42]. *Continuity of $\mathcal{T}$* follows from Lipschitz continuity of f and $\tilde{p}$ in μ (R1). *Relative compactness* follows from Prokhorov's theorem under

(R2) and (R4). By Schauder's theorem, $\mathcal{T}$ has a fixed point $\mu^* = \mathcal{T}(\mu^*)$; setting $\Phi^* = \Phi(\cdot;\mu^*)$ yields the MFE. ■

Theorem 3 (MFG Approximation of Nash Equilibrium) *Let Assumptions 2 and 3 hold, and let (Φ^*, μ^*) be a MFE. Define the MFE-induced strategy profile:*

$$b_i^{N,*} = b^*(\mathbf{s}_i, t; \mu^*), \quad i = 1, \dots, N. \tag{46}$$

Then $\mathbf{b}^{N,*} = (b_1^{N,*}, \dots, b_N^{N,*})$ is an ε_N-Nash Equilibrium (Definition 5) of the PAG, where:

$$\varepsilon_N = \frac{2L_u C}{\sqrt{N-1}} = \mathcal{O}\left(\frac{1}{\sqrt{N}}\right), \tag{47}$$

and $C > 0$ depends only on the second moment of μ^* and the dimension of $\mathcal{S}$.

Proof: Fix arbitrary $i \in \mathcal{N}$ and $\hat{b}_i \in \mathcal{B}$.

Step 1 (Empirical measure decomposition). States $\{\mathbf{s}_j\}_{j\neq i}$ are i.i.d. from μ_0^* (MFE self-consistency). Define $\mu_{-i}^N = \frac{1}{N-1}\sum_{j\neq i}\delta_{\mathbf{s}_j}$. By the Wasserstein concentration inequality [43]:

$$\mathbb{E}\big[W_1(\mu_{-i}^N, \mu_0^*)\big] \le \frac{C}{\sqrt{N-1}}. \tag{48}$$

Step 2 (Utility gap decomposition).

$$\begin{aligned} &u_i(\hat{b}_i, b_{-i}^{N,*}) - u_i(b_i^{N,*}, b_{-i}^{N,*}) \\ &= \underbrace{u_i(\hat{b}_i, b_{-i}^{N,*}) - u^{\mathrm{MFG}}(\mathbf{s}_i, \hat{b}_i; \mu_0^*)}_{\text{(I)}} \qquad (49) \\ &\quad + \underbrace{u^{\mathrm{MFG}}(\mathbf{s}_i, \hat{b}_i; \mu_0^*) - u^{\mathrm{MFG}}(\mathbf{s}_i, b_i^{N,*}; \mu_0^*)}_{\text{(II)}} \\ &\quad + \underbrace{u^{\mathrm{MFG}}(\mathbf{s}_i, b_i^{N,*}; \mu_0^*) - u_i(b_i^{N,*}, b_{-i}^{N,*})}_{\text{(III)}}. \qquad (50) \end{aligned}$$

Step 3 (Bound on (II)). Since $b_i^{N,*} = b^*(\mathbf{s}_i; \mu_0^*)$ maximises $u^{\mathrm{MFG}}(\mathbf{s}_i, \cdot\,; \mu_0^*)$: $(\mathrm{II}) \le 0$.

Step 4 (Bound on (I) and (III)). Using Assumption 3: $|(\mathrm{I})| + |(\mathrm{III})| \le 2L_u W_1(\mu_{-i}^N, \mu_0^*)$.

Step 5 (Conclusion). Taking expectations and applying the concentration bound:

$$\mathbb{E}\Big[u_i(\hat{b}_i, b_{-i}^{N,*}) - u_i(b_i^{N,*}, b_{-i}^{N,*})\Big] \le \frac{2L_u C}{\sqrt{N-1}} = \varepsilon_N. \tag{51}$$

Since this holds for all i and all $\hat{b}_i$, $\mathbf{b}^{N,*}$ is an ε_N-Nash Equilibrium. ■

Proposition 1 (Variance Reduction via Value Baselining) *Let $X_i^{(r)} := u_i(b_i'^{(r)}, b_{-i}; b'_{\mathrm{MFG}}, \theta) - u_i(b_i, b_{-i}; b_{\mathrm{MFG}}, \theta)$ denote the raw deviation gain at PGA step r, and let $\hat{X}_i^{(r)} := A_i(b_i'^{(r)}, b'_{\mathrm{MFG}}; \theta) - A_i(b_i, b_{\mathrm{MFG}}; \theta)$ denote the baselined variant. For any baseline c independent of b'_i, $\mathbb{E}[\hat{X}_i^{(r)}] = \mathbb{E}[X_i^{(r)}]$, so the regret estimator is unbiased. Moreover, the baseline $c^* = \widehat{\Phi}_\xi(b_i, b_{\mathrm{MFG}})$ minimises $\mathrm{Var}(\hat{X}_i^{(r)})$ in the class of state-dependent baselines, reducing variance by $\mathrm{Var}(\widehat{\Phi}_\xi)/\mathrm{Var}(X_i^{(r)})$. Consequently, when $\widehat{\Phi}_\xi$ closely approximates the MFE value Φ^*, the baselined gradient estimate has lower variance than the raw estimate, yielding faster convergence of the augmented Lagrangian updates.*

Proof Sketch: Unbiasedness follows because $\mathbb{E}_{b'_i}[c] = c$ is constant w.r.t. the deviation distribution, so subtracting it preserves the expectation. The variance-minimising baseline is the conditional expectation $c^* = \mathbb{E}[X_i^{(r)} | b_i, b_{\mathrm{MFG}}]$, which equals the MFE value $\Phi^*(s(b_i), \mu^*)$ when b^* is the equilibrium strategy. The variance reduction formula follows from the identity $\mathrm{Var}(X - c^*) = \mathrm{Var}(X) - \mathrm{Var}(c^*)$. ■